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Lecture 6: Introduction to Data Modelling

COMP1037 Fundamentals of Artificial Intelligence

Xinan Chen

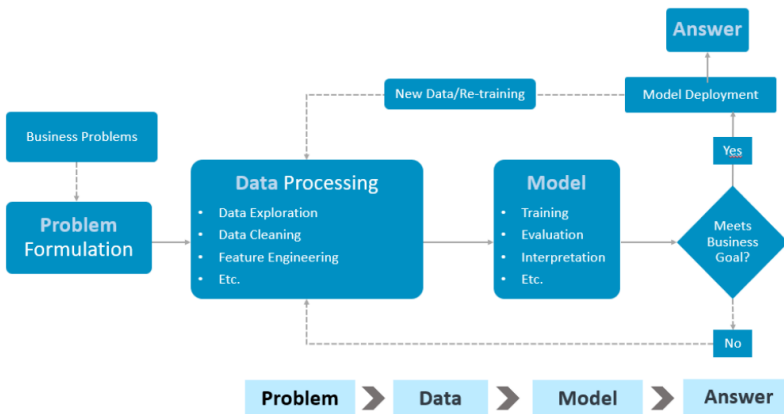
University of Nottingham Ningbo China

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Outline |

1. Data Analysis and Modelling
2. Data
3. Data Analysis
4. Data Modelling
5. Model Evaluation and Validation

The Data Pipeline |



© AWS Machine Learning



Definition of Data Analysis and Modelling |

Data Analysis

The process of **examining, cleaning, transforming, and interpreting data** to extract useful insights and support decision-making.

Data Modelling

The creation of abstract representations (models) to describe, analyze, and predict real-world phenomena.

Relationship Between Analysis and Modelling

- Data analysis provides the foundation for modelling by identifying **patterns, trends, and relationships** in the data.
- Modelling uses these insights to create predictive or descriptive models.

Research Questions and Negotiation |

The analysis starts with the **needs and objectives** from stakeholders.

You need to understand the requirements of the analysis:

- Who is it for (at what level are you pitching it)? Professors, manager?
- What resources do you have? Existing dataset/new data?
- How much time is available?
- How will the analysis be validated?
- What level of uncertainty is acceptable?
- Does it need to be reproducible? (Less obvious than you think!)

Those things need to be **negotiated** with the stakeholders.

Do not assume stakeholders know what they need.

Characteristics of a Good Analysis Question |

The SPAIN Characteristics:

- Specific:**
- “Does eating vegetables improve health?” (Too general)
 - “Does increasing vegetable consumption improve blood sugar in diabetics?” (Specific)

- Plausible:**
- “Does eating vegetables increase your bench press?” (No plausible link)
 - “Does your credit limit influence your salary?” (Causality link is reversed)

- Answerable:**
- Is it possible to collect the data to answer the question?
 - Does the data even exist?

Interesting to a Specific Audience: Who is your audience?
(Researcher, boss, friends)

Novel: Has someone else recently answered it? Nobody likes to do work for no reason.

The Epicycle of Analysis |

Main Activities of Data

Analysis:

1. Stating the question
2. Exploring and analysing the data
3. Building a model
4. Interpreting the results
5. Communicating the results

Sub-Activities for Each:

1. Setting expectations
2. Collecting data/information and comparing it to expectations
3. Revising expectations

Source: Peng, R.D., & Matsui, E. (2015). *The Art of Data Science: A Guide for Anyone Who Works with Data*.

Describing the Data |

- **Stevens' four level scale:**

- Nominal
- Ordinal
- Interval
- Ratio

- **Mosteller and Tukey's scale:**

- Names
- Grades
- Ranks
- Fractions
- Counts
- Amounts
- Balances

- **Chrisman's expanded scale:**

- Nominal
- Gradation of membership
- Ordinal
- Interval
- Log-interval
- Extensive ratio
- Cyclical ratio
- Derived ratio
- Counts
- Absolute

Describing the Data (continued) |

- Nominal
 - Names, labels, categories, etc.
 - Unordered and distinct
 - Relevant statistics: **mode**
 - Bar charts as far as the eye can see
- Ordinal
 - Scales, ratings, etc.
 - Ordered
 - Distinct (no overlap)
 - Relevant statistics: **median, percentile**
- Interval
 - Numeric
 - Ordered, but differences are meaningful
 - Distinct, ordered and **additive**
 - Interval data has **no meaningful 0**
 - Relevant statistics: **mean, standard deviation, correlation, regression**
- Ratio
 - Numeric
 - Ordered and can be used with any arithmetic operator
 - Distinct, ordered, additive, multiplicative
 - Ratio data **has a meaningful, absolute 0**
 - **All statistics apply**

Describing the Data - Examples |

1. Nominal Data

Concept: Categories only. You can count frequencies, but cannot rank or do math.

ID	Blood	OS
001	O	Win
002	A	Mac
003	AB	Linux
004	B	Win

2. Ordinal Data

Concept: Logical rank/order, but exact distance between ranks is unknown.

ID	Rating	Income
884	*****	High
885	**	Low
886	***	Med
887	*****	Med

Describing the Data - Examples |

3. Interval Data

Concept: Meaningful differences, but **no true zero** (e.g., 0°C isn't "no temp").

ID	Year	Temp ($^{\circ}\text{C}$)
A1	1990	-5.0
A2	2005	2.5
A3	1985	0.0
A4	1999	10.0

4. Ratio Data

Concept: A **true zero** exists. All math operations are valid (e.g., twice as heavy).

ID	Area (m^2)	Price
H1	120.5	\$450k
H2	85.0	\$310k
H3	240.0	\$890k
H4	0.0	\$50k

Describing Data |

We use numbers and drawings to get a sense of a large dataset.

Location: What's the most common value?

Variability: How spread out is the data? What's a deviant value (outlier)?

Shape: How is the data distributed?

Relationship: How are two variables linked?

Summary Statistics – Central Tendency |

Mode: Most frequent attribute in the sample. Not necessarily unique.

Median: Middle value in a sorted dataset.

■ If n is odd: Median = $x_{\frac{n+1}{2}}$

■ If n is even: Median = $\frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$

Arithmetic Mean: $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

Summary Statistics – Central Tendency |

Scenario: Let's look at the annual salaries of 5 people at a small tech startup.

\$40k \$45k **\$50k** \$50k **\$400k (CEO)**

The Calculations

- **Mode:**
- **Median:**
- **Mean:**

Summary Statistics – Central Tendency |

Scenario: Let's look at the annual salaries of 5 people at a small tech startup.

\$40k \$45k **\$50k** \$50k **\$400k (CEO)**

The Calculations

- **Mode:** \$50k (*Appears most often*)
- **Median:** \$50k (*The exact middle value*)
- **Mean:** \$117k (*Sum $\div$ 5*)

The Takeaway

- The **Mean (\$117k)** is dragged drastically upward by the CEO's salary. It does not represent a "typical" worker.
- The **Median (\$50k)** completely resists the outlier, accurately reflecting the reality of the standard employee.

Summary Statistics – Variability |

Range:

Difference between minimum and maximum value. Allows you to quickly see the bounds of the data.

Inter-Quartile Range (IQR):

The p -quantile of data is the value relative to which the fraction p of the data is smaller.

The $\frac{1}{2}$ -quantile value is the median.

The p -inter-quantile range is the difference between the $(1 - p)$ -quantile value and the p -quantile value.

It is common to use $p = \frac{1}{4}$ (inter-quartile range).

- Average absolute deviation

$$d_{\bar{x}} = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$$

- Variance

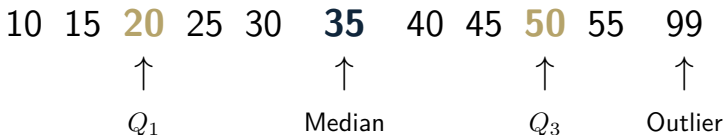
$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Standard deviation

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Understanding Inter-Quartile Range (IQR) |

Scenario: Let's find the IQR for a sorted dataset of 11 exam scores.



Applying the Formula

- **p -quantile (Q_1 , bottom 25%):** The value where a quarter of the data is smaller → **20**
- **$(1 - p)$ -quantile (Q_3 , top 25%):** The value where three-quarters of the data is smaller → **50**
- **IQR:** $Q_3 - Q_1 = 50 - 20 = 30$

Why use IQR instead of the Total Range? |

Recall our dataset: 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, **99**

1. The Total Range

- **Formula:** $\text{Max} - \text{Min}$
- **Calculation:** $99 - 10 = 89$
- **The Problem:** The total range is **89**, but almost all students scored between 10 and 55. The single extreme score of 99 drastically distorts our view of the data's spread.

2. The IQR

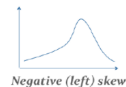
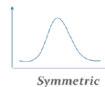
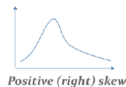
- **Formula:** $Q_3 - Q_1$
- **Calculation:** $50 - 20 = 30$
- **The Advantage:** By only measuring the spread of the **middle 50%** of the data, the IQR completely ignores the outlier, giving a much more accurate and robust picture of normal variance.

Summary Statistics – Shape (Skewness) |

Skewness:

Measures the asymmetry of the **probability distribution** of a real-valued random variable about its mean.

- **Positive (Right) Skew:** Tail is longer on the right.
- **Symmetric:** No skew.
- **Negative (Left) Skew:** Tail is longer on the left.

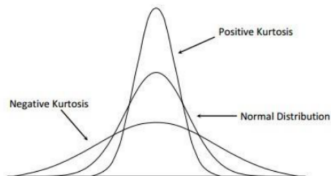


Summary Statistics – Shape (Kurtosis) |

Kurtosis:

Measures how peaked a distribution is.

- **Mesokurtic:** Looks like a Gaussian (normal) distribution.
- **Platykurtic:** Wider and flatter than a Gaussian (negative kurtosis).
- **Leptokurtic:** Thinner and peakier than a Gaussian (positive kurtosis).



Measure is usually computed relative to the Gaussian distribution.

Summary Statistics – Relationships |

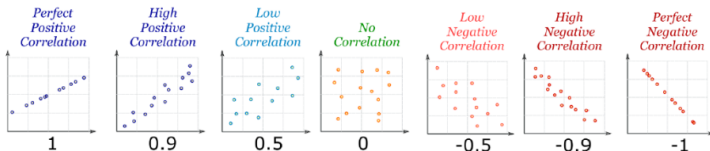
Pearson Correlation Coefficient:

Measures the strength and direction of the linear relationship between two variables. Ranges between $[-1, +1]$:

- -1 : Perfect negative linear relationship.
- 0 : No linear relationship.
- $+1$: Perfect positive linear relationship.

Formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$



Diagnosing Data Problems |

Check for Boundary Violations

Impossible Values: If a field contains “Age,” negative values are impossible.

Outliers: Different from impossible values!

Example: An “Age” value of 185.

Knowing how the data is generated helps in identifying faulty data.

Check the Packaging

- Look at the first few rows and the last few rows. Does it all look as expected?
- Check the dimensions of your dataset (number of observations, number of columns). Does it all match up?

Diagnosing Data Problems – Imputation |

Imputation: Replacement of missing data with substituted values.

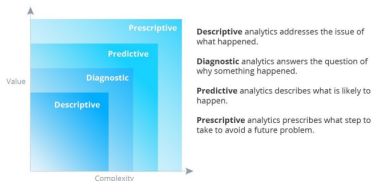
- **Unit Imputation:** Impute a data point (e.g., remove a person with missing Age).
- **Item Imputation:** Impute a value of a data point (e.g., replace missing Age value).
- **Hot Deck Imputation:** Replace value by value of previous observation.
- **Cold Deck Imputation:** Replace value by value randomly sampled from an external dataset.
- **Mean Substitution:** Replace value by central value of all other cases (mean/mode/median).
- **Regression Imputation:** Use a basic machine learning model to predict the missing value.
- **Multiple Imputations:** Sometimes you need more than one method! (depends on data)

Overview of Data Analysis Techniques |

Purpose: To transform raw data into **actionable insights**.

Types of Data Analysis:

1. Descriptive Analysis
2. Diagnostic Analysis
3. Predictive Analysis
4. Prescriptive Analysis
5. Exploratory Data Analysis (EDA)



Source: Things You Should Know About Types Of Data Analysis

Descriptive Analysis |

Definition

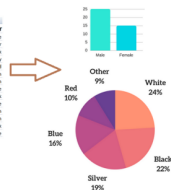
Summarizes historical data to understand what has happened.

Examples:

- Mean, median, mode.
- Sales reports, customer demographics.

	A	B	C	D
1	Respondent Number	Age	Gender	Favorite Car Color
2	1	22	M	White
3	2	37	F	Silver
4	3	45	F	Black
5	4	62	F	Gray
6	5	28	M	Red
7	6	45	M	Green
8	7	88	F	Brown
9	8	65	M	White
10	9	95	M	Black
11	10	27	M	White
12	11	38	F	Green
13	12	43	M	Brown
14	13	55	F	Black
15	14	58	F	White

RAW DATA



Descriptive Statistics

Source: Descriptive Statistics
Examples, Types and Definition

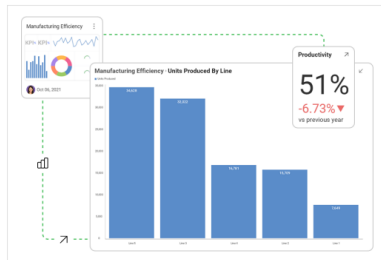
Diagnostic Analysis |

Definition

Identifies the causes of past events or behaviors.

Examples:

- Root cause analysis.
- Correlation analysis.



Source: Types of Data Analysis, Benefits & Examples

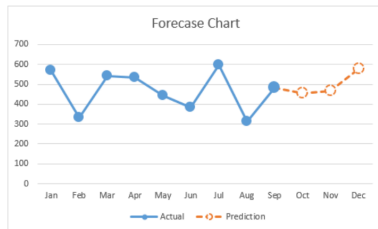
Predictive Analysis |

Definition

Uses historical data to predict future outcomes.

Examples:

- Sales forecasting, customer churn prediction.



Source: Forecast Chart

Prescriptive Analysis |

Definition

Recommends actions based on data insights.

Examples:

- Optimization models, decision trees.



Source: Types of Data Analysis, Benefits & Examples; Data Analysis 101

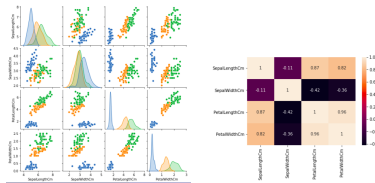
Exploratory Data Analysis (EDA) |

Definition

Investigates data to discover patterns, trends, and anomalies.

Examples:

- Pairplots, heatmaps, summary statistics.



Source: Exploratory Data Analysis on Iris Dataset

What is a Model? |

Definition

A simplified representation of a real-world system.

Purpose

To understand, predict, or simulate real-world phenomena.

Types of Models |

- 1. Statistical Models:** Uses probability and statistical techniques to make inferences from data.
Examples: Regression models, time-series models.
- 2. Machine Learning Models:** Uses data-driven algorithms to learn patterns and make predictions.
Examples: Decision Trees, Neural Networks.
- 3. Simulation Models:** Mimics real-world processes using computational techniques.
Examples: Monte Carlo simulations, agent-based models.

Introduction to Statistical Models |

Definition

A statistical model is a mathematical representation of relationships between variables in data.

- Used for **inference**, **prediction**, and **hypothesis testing**.
- Helps in understanding patterns and making data-driven decisions.
- Examples: Regression models, time-series models, probabilistic models.

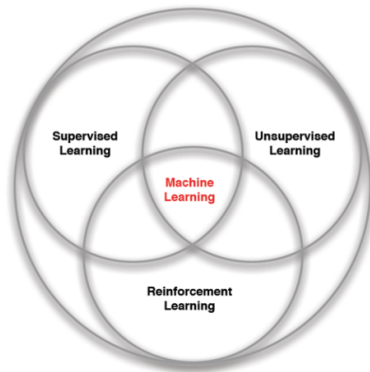
Introduction to Machine Learning Models |

Definition

Models that learn patterns from data.

Types:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

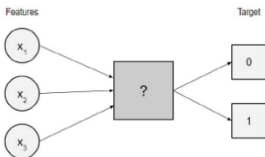


Supervised Learning |

Definition

A machine learning approach where the model is trained using **labeled data**.

- Input (features) and output (targets/labels) are known.
- Data: (x, y) x is data, y is target/label.
- Learn a function to map $x \rightarrow y$.
- Applications: Classification, regression, object detection, semantic segmentation, image captioning, etc.
- Examples: Linear regression, logistic regression, decision trees, naïve Bayes, neural networks, and many more.

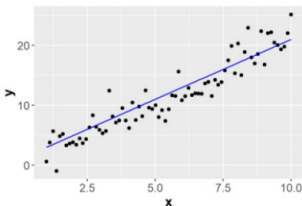


Features x				Target y
Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.3	3.0	1.1	0.1	setosa
5.0	3.3	1.4	0.2	setosa
7.7	3.8	6.7	2.2	virginica
5.5	2.5	4.0	1.3	versicolor

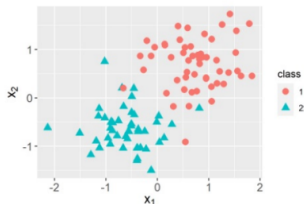
Supervised Learning – Regression vs. Classification |

The two most basic supervised learning models are **regression** and **classification**, depending on whether the target is numerical or categorical.

Regression: Our observed labels come from $Y \subseteq \mathbb{R}$.



Classification: Observations are categorized: $y \in Y = \{C_1, \dots, C_g\}$.

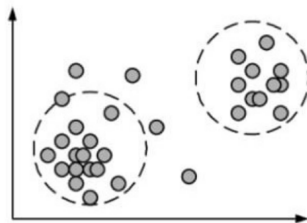


Unsupervised Learning |

Definition

A machine learning approach where the model learns patterns **without labeled data**.

- Identifies hidden structures and relationships in data.
- x – Just data, no labels!
- Examples: K-means clustering, hierarchical clustering, principal component analysis (PCA), and many more.



Source: SuperAnnotate

Reinforcement Learning |

Definition

A learning paradigm where an agent learns by interacting with an environment and receiving rewards or penalties.

- Used in robotics, gaming, and autonomous systems.
- Learn how to take actions in order to maximize reward.
- Key components: **Agent**, **Environment**, **Reward**, **Policy**.



Source: Reinforcement Learning

Simulation Models |

Definition

A model that uses computational techniques to simulate real-world processes and assess outcomes.

- Example: Monte Carlo simulation for risk assessment.
- Used in finance, healthcare, and engineering.



Source: An Introduction and Step-by-Step Guide to Monte Carlo Simulations

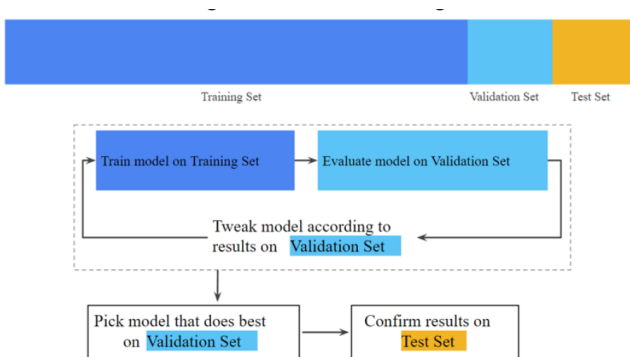
Importance of Model Evaluation |

Why Model Evaluation Matters:

- Ensures model reliability and robustness.
- Helps detect **overfitting** and **underfitting**.
- Assists in selecting the best model for a given task.
- Provides insights into model performance on unseen data.

Train-Test Split / Holdout Method |

A simple validation technique where data is split into **training**, **validation**, and **testing** sets.



Source: Pi.Exchange

Cross-Validation |

Cross-Validation

A more robust method where the dataset is divided into **multiple folds**.

k-Fold Cross-Validation

Data is split into k subsets, and each subset is used for testing once. Helps improve model generalization.

Iteration 1	Test	Train	Train	Train	Train
Iteration 2	Train	Test	Train	Train	Train
Iteration 3	Train	Train	Test	Train	Train
Iteration 4	Train	Train	Train	Test	Train
Iteration 5	Train	Train	Train	Train	Test

Model Evaluation Metrics |

Key Metrics for Regression:

- R^2 (Coefficient of Determination)
- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)

Key Metrics for Classification:

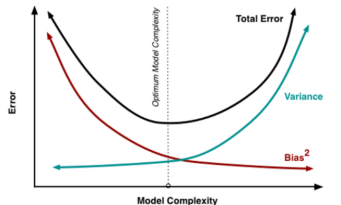
- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Receiver Operating Characteristic (ROC) Curve
- Area Under the Curve (AUC)

Bias-Variance Tradeoff |

Bias: Error due to overly simplistic assumptions.

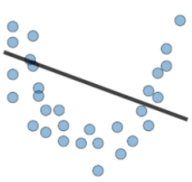
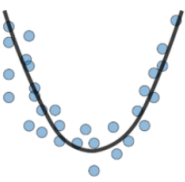
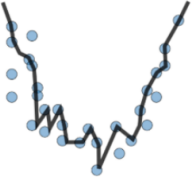
Variance: Error due to excessive sensitivity to training data.

- High bias leads to **underfitting**.
- High variance leads to **overfitting**.
- The goal is to find an optimal balance between bias and variance.



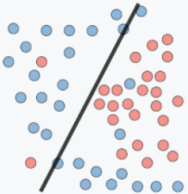
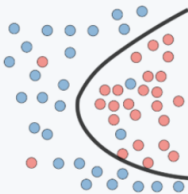
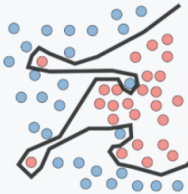
Source: Dataquest

Underfitting vs Overfitting |

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Regression illustration			

Source: Machine Learning tips and tricks cheatsheet

Underfitting vs Overfitting (cont.) |

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Classification illustration			

Source: Machine Learning tips and tricks cheatsheet

Underfitting vs Overfitting (cont.) |

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Deep learning illustration	 The graph shows Error on the y-axis and Epochs on the x-axis. A red line (Training) and a blue line (Validation) both start high and decrease slightly, remaining very close to each other and at a high level.	 The graph shows Error on the y-axis and Epochs on the x-axis. A red line (Training) starts high and decreases to a low level. A blue line (Validation) starts high and decreases to a level slightly above the training error.	 The graph shows Error on the y-axis and Epochs on the x-axis. A red line (Training) starts high and decreases rapidly to near zero. A blue line (Validation) starts high, decreases to a minimum, and then increases significantly.

Source: Machine Learning tips and tricks cheatsheet

Underfitting vs Overfitting (cont.) |

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Possible remedies	• Complexify model • Add more features • Train longer		• Perform regularization • Get more data

Source: Machine Learning tips and tricks cheatsheet

Characteristics of a Good Model |

Accuracy: Produces results that match real-world observations.

Generalizability: Works well with unseen data.

Interpretability: Can be easily understood and explained.

Efficiency: Uses computational resources effectively.

Scalability: Performs well with increasing data.

Model Selection Criteria |

Performance Metrics: Accuracy, RMSE, Precision, Recall.

Interpretability: How easily the model's decisions can be explained.

Computational Efficiency: Resource usage and execution time.

Generalization Ability: How well the model performs on unseen data.

Data Availability: The amount and quality of data required for training.